

RAJALAKSHMI ENGINEERING COLLEGE

CR23532 – CYBER AND DIGITAL FORENSICS

Case Study Assignments – II (Answers)

Submitted by

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ASSIGNMENT 2(1)

Suspected Unauthorized Access to a College Server

Scenario selected: Confidential semester exam question papers were copied from the Computer Science department server between 11 PM and 1 AM. The server log shows the account of staff member "User-X". A USB drive was found in User-X's workstation.

Assumptions: The college has a written IT policy, the administrator gave written permission to investigate, the server clock is synchronised with NTP, and the workstation was powered off when the team arrived.

Q1. How the team moves through the five phases

A digital forensic investigation follows a fixed order so that evidence is not lost or damaged and every step can be explained later. In this case the phases work as follows.

1. Identification. The team first records the administrator's complaint: what was accessed, when it was noticed, and who reported it. They then decide what could hold evidence (server, server logs, suspect workstation, USB drive, the exam files) and what the investigation must prove (who accessed the files, when, how, and where they were copied to). Written authorization is obtained before anything is touched. A case number is created.

2. Collection. Evidence is secured and gathered without changing it. The workstation and USB are photographed, labelled and sealed. Disks are imaged through a write blocker, and the server logs are copied with their hash values. Volatile data such as running processes is captured first if the server is still running. Every transfer is entered in the chain-of-custody form.

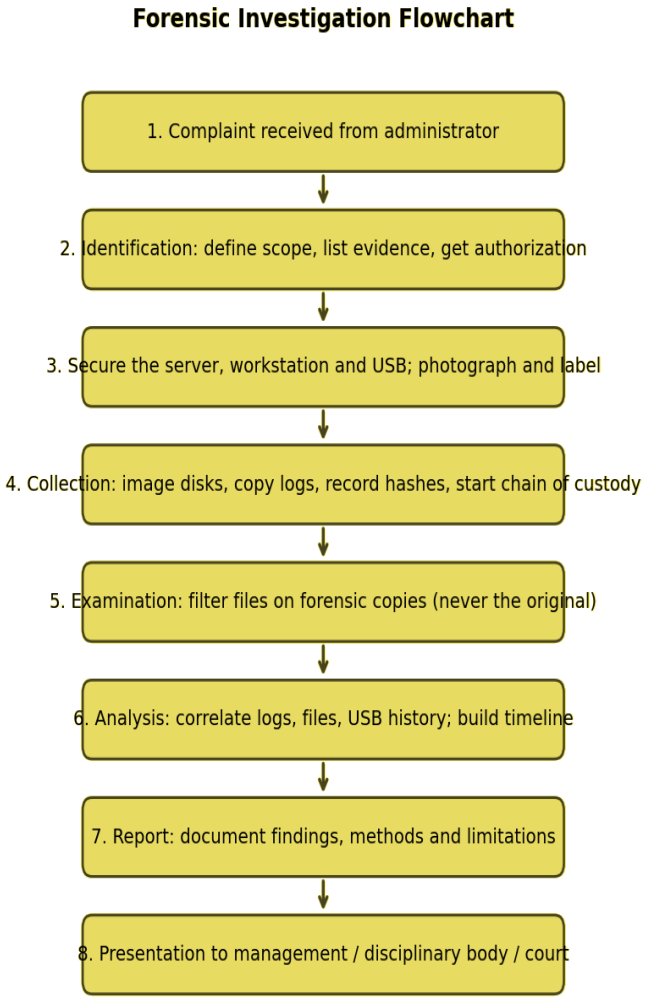
3. Examination. The team works only on the forensic copies. They filter out irrelevant data, and extract what matters: the exam files, login records, recently opened files, USB connection history and deleted files that can be recovered.

4. Analysis. The extracted data is studied and linked together. Server logs show which account logged in and from which IP; the workstation shows when the USB was connected; the USB shows which exam files were copied. Putting these on one timeline shows whether User-X really did it, or whether someone else used the account.

5. Presentation. The findings are written in a report and explained to the management, disciplinary committee or court. The report shows what was found, how it was found, and what the limits of the evidence are.

Q2. Investigation flowchart

The flowchart below shows the sequence from the first complaint to the presentation of findings.



Q3. Potential digital evidence and its investigative value

Evidence is classified by how strongly it helps the investigation. Direct evidence shows the action itself. Supporting evidence gives context, and lead evidence only points the team in a direction.

No.	Evidence	What it shows	Value
1	Server access logs	Login time, account name, source IP address, files opened and copied	High (direct)
2	USB device	Copies of the exam files, volume serial number and file times on the device	High (direct)
3	Suspect workstation (disk image)	USB connection history in the registry, recently opened files, deleted files, login times	High (direct)
4	Exam files on the server	Created, modified and last-accessed times show which files were touched	Medium (shows what was affected)
5	Network / firewall logs	Large data transfer or connection from the workstation at the same time	Medium (supporting)
6	CCTV / door-entry records	Whether User-X was physically in the building at that time	Medium (supporting)
7	Administrator's complaint	Starting point and time of discovery	Low (lead only)

Q4. Precautions during collection and preservation

The aim is that the evidence stays unchanged and that anyone can later check this. The main precautions are:

- Authorization: get written permission before seizing devices. Evidence taken without it may be rejected.
- Photograph and record: take photographs of the scene, cables, screen and device state (on/off). Note serial numbers. This shows how the evidence looked when found.
- Volatile data first: if the server is running, capture memory, running processes and network connections before shutting down, because this data is lost on power-off.

- Write blocker and imaging: connect the disk through a write blocker and take a bit-by-bit forensic image. Never examine the original.
- Hashing: record MD5/SHA-256 values of the original and the copy. Matching values prove nothing changed.
- Packing and storage: use anti-static bags and evidence seals, and store in a locked room away from heat, magnets and moisture.
- Chain of custody: log who handled the evidence, when and why. A gap in this log can make the evidence doubtful.
- Documentation: write down every action, tool and version used, with date and time.

Q5. Establishing the timeline of the unauthorized access

A timeline is built by collecting time stamps from every source, converting them to one time zone, and placing them in order. The sources are:

- Server logs: login and logout time, IP address, files opened and copied.
- File system metadata: created, modified and accessed times of the exam files on the server.
- Workstation artifacts: Windows event logs (logon times), registry entries that show when the USB was first and last connected, and recent-file lists.
- USB device: file creation times on the drive, which show when the copy happened.
- Network and CCTV records: connect the activity with the location and with the person.

Before using the times, the examiner checks that the server and workstation clocks were correct and notes any difference. The sample timeline below shows how the pieces fit together.

Time	Event	Source
22:58	User-X's account logs in to the server from the workstation IP	Server log
23:05	Exam folder opened	Server log / file system
23:10	USB drive connected to the workstation	Registry (USBSTOR)
23:12	Exam files copied from the server to the USB	File times / USB
23:30	USB removed; user logs out	Registry / server log
08:30 (next day)	Administrator notices and reports the incident	Complaint

Q6. Final forensic report and presenting to a non-technical authority

The report should be clear and complete enough that another examiner could repeat the work. A sample structure:

1. Title page and case details (case number, date, examiner, requester).
2. Executive summary: a short, plain-language summary of what happened and what was found.
3. Scope and authorization: what was examined and under whose permission.
4. Evidence list: every item, with serial numbers and hash values.
5. Tools and methods: tools, versions and the procedure followed.
6. Findings: facts found, with the timeline and supporting screenshots.
7. Analysis and conclusion: what the findings mean, and what is not proven.
8. Limitations and assumptions.
9. Appendices: chain-of-custody forms, log extracts, glossary.

For a non-technical authority such as the principal or a committee: start with a short summary, use plain words and explain technical terms in one line (for example, a hash is a digital fingerprint of the data). Show a simple timeline chart, give facts instead of opinions, and clearly separate what is proven from what is only likely.

Q7. Legal and ethical considerations in a college environment

- Authorization and policy: the college's acceptable-use policy and written management approval permit the examination of college-owned systems. Personal devices need the owner's consent or legal authority.
- Privacy: staff and students keep a reasonable right to privacy. Collect only what is relevant to the case and do not read unrelated personal emails or files.

- Law: the IT Act 2000 (for example Sections 43 and 66 on unauthorized access and data theft) applies. Electronic records are admitted under the Indian Evidence Act / Bharatiya Sakshya Adhiniyam, which needs a certificate and proof of integrity.
- Neutrality: the examiner must report facts honestly, even if they do not support the complaint, and must not be biased against the suspect.
- Confidentiality: the exam data and the findings must be shared only with authorized people.
- Admissibility: proper chain of custody, hashing and documentation keep the evidence usable in a disciplinary hearing or court.

ASSIGNMENT 2(2)
Linking a Suspicious File to a User Using File-System Evidence

Scenario selected: A shared Windows 10 office desktop contains a suspicious file "update_tool.exe", several hidden files and encrypted documents. User accounts: Admin, Ravi, Meena and Guest. Investigators suspect that the account "Ravi" created and used the files.

Assumptions: The disk was imaged correctly, the file system is NTFS, and ILookIX is used for analysis.

Q1. Hardware and software to document before analysis

Documenting the system first records exactly what was examined, helps explain the evidence later, and shows that nothing was added or changed. The following items are recorded:

Hardware	Software
Make, model and serial number of the computer	Operating system, version and installation date
Hard disk / SSD model, size and serial number	File system (NTFS, FAT32) and partitions
RAM, motherboard, network card	Installed applications, browsers and email clients
Connected devices and USB ports	User accounts, groups and passwords policy
BIOS date and time (compared with real time)	Security tools, antivirus and encryption software
Physical condition and photographs	Time zone and system clock settings

Q2. How storage, operating systems, file systems and metadata help locate evidence

- Storage devices: the disk layout (partitions, unallocated and slack space) tells the examiner where to search. Deleted or hidden data often remains in unallocated space.
- Operating system: Windows keeps user profiles, the registry, event logs, Prefetch, recent-file lists and the Recycle Bin. These show which user ran which program and when.
- File system: NTFS stores file name, path, size, owner and time stamps in the Master File Table ($MFT). Even deleted files may still have entries, so the examiner can see that they existed.
- Metadata: data about data, such as author, creation time, last-modified time, last-accessed time and file owner. It helps place a file in time and tie it to a user account.

Q3. Effect of hidden and encrypted files, and evidence-handling considerations

Hidden files: A hidden file is not shown by default, and a user may hide files to avoid being noticed. Hidden files can be marked with the hidden or system attribute, or placed in obscure folders. They may hold evidence, so the examiner must display hidden and system files and list them separately. Hiding does not by itself prove guilt, because many system files are hidden normally.

Encrypted files: Encryption makes the content unreadable without the key or password. The investigation is slowed or blocked, but the examiner can still see the file name, size, path and time stamps. Keys, passwords or recovery keys may be found elsewhere on the system, such as saved passwords, notes, or the encryption program's settings.

Handling: Work only on the image and keep the original files unchanged. Record hash values of the encrypted files. Do not try to break passwords without legal authority. Note in the report which files could not be opened and why. Keep the whole process documented.

Q4. Evidence-linkage table

Each artifact alone is weak. The strength comes from several artifacts pointing to the same user. The table shows how they are correlated.

Artifact	Example	How it links to the user
File name / path	C:\Users\Ravi\Downloads\update_tool.exe	The path lies inside Ravi's profile folder
Timestamps	Created 10:02, last run 10:05 on 14-Oct	Match the time Ravi was logged in
User account / SID	File owner = Ravi (SID ending 1004)	Ownership points to Ravi's account
File metadata	Author / last-saved-by = Ravi	Shows who created or edited the document
Browser artifacts	Download record of update_tool.exe in Ravi's Chrome profile	Shows how the file arrived and who downloaded it
Event / login logs	Ravi logged in 09:55 and logged out 11:20	Places Ravi's account at the computer
Prefetch / UserAssist	update_tool.exe run count = 2	Proves the program was executed, under Ravi's profile
Hidden / encrypted files	Hidden folder in Ravi's Documents	Stored in the same user area

Q5. Workflow for importing and analysing the evidence in ILookIX

1. Create a new case in ILookIX and enter the case number, examiner name and description.
2. Add the evidence: import the forensic image, or acquire the disk using a write blocker. Verify the hash value matches the acquisition record.
3. Let ILookIX process the image: it reads the partitions and file system, lists files (including deleted and hidden), and builds an index for fast search.
4. Review the user accounts and profiles to see which users exist and how active each is.
5. Use filters and keyword search to find the suspicious .exe, hidden files, encrypted documents and browser data. Use hash sets to ignore known system files.
6. Check the path, time stamps and metadata of each item and compare them with login records.
7. Bookmark the relevant items and add notes. Build a timeline of activity.
8. Export a report with bookmarks, hash values and the case notes.

Q6. Direct artifact versus supporting artifact

A direct artifact ties a specific user to a specific action. A supporting artifact only gives background, and cannot prove who did the action by itself.

Direct (links user to activity)	Supporting / contextual
File created in Ravi's own profile while only Ravi was logged in	The computer has several user accounts
Download record in Ravi's browser profile for this exe	The file exists on the system
Prefetch / UserAssist entry showing Ravi's account ran the program	Hidden attribute set on a file
Document metadata showing Ravi as author	Windows installation date or installed software list

Q7. Limitations and alternative explanations

- Shared computer: another staff member may have used Ravi's login session, or the password may be known to others. So the account is not the same as the person.
- Malware or remote access: the files might have been created by malware or by a remote user without Ravi's knowledge.
- Time problems: a wrong system clock, time-zone differences, or copying and moving files can change time stamps and give a wrong sequence.
- Automatic activity: the file may have come from an automatic update, a pushed installer or an email attachment, and Ravi may not have opened it knowingly.
- Incomplete data: cleared logs, deleted history or encrypted content may hide the full picture.

ASSIGNMENT 2(3)
Forensic Imaging and Safekeeping of a Suspect Storage Device

Scenario selected: A Dell laptop with a 512 GB SSD is seized from an employee suspected of copying client data. A 64 GB USB drive was plugged into the laptop at the time of seizure.

Assumptions: A search authorization is in place, the laptop is Windows-based, and a hardware write blocker and IXImager are available in the lab.

Q1. Safe collection and transport procedure

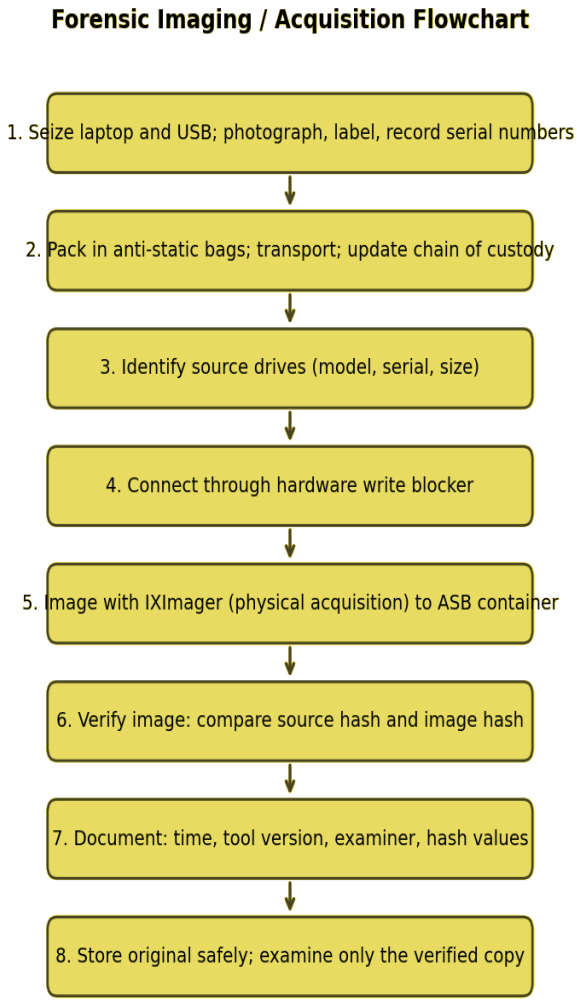
1. Secure the area and allow nobody to touch the devices. Take photographs of the laptop, USB, screen, cables and surroundings.
2. If the laptop is on, note the screen and capture volatile data (RAM, open programs, network connections). Then shut down according to lab policy. Do not open or browse files.
3. Carefully remove the USB drive and, where possible, the laptop battery to prevent remote changes.
4. Label each item with the case number, date, time, location and your name. Note make, model and serial numbers.
5. Place the items in separate anti-static bags and seal them with tamper-evident tape.
6. Transport in a padded case, away from heat, magnets and moisture. Do not leave the evidence unattended.
7. Complete the chain-of-custody form at every handover until the evidence reaches the lab.

Q2. Purpose of physical acquisition and why a forensic image is used

Physical acquisition copies every sector of the storage device, bit by bit, regardless of the file system. This includes deleted files, unallocated space, slack space and hidden areas that normal copying would miss.

- Protecting the original: opening or booting the original changes time stamps and system files, and may overwrite deleted data. Using an image avoids this.
- Repeatability: if the image is damaged or a different tool is needed, another copy can be made from the untouched original.
- Integrity: the image can be verified by hash value, so the court can trust that it is identical to the original.
- Safety: the original stays safely sealed, so a mistake during analysis does not destroy the evidence.

Q3. Imaging workflow using IXImager



- Source identification: record the model, serial number and capacity of the SSD and the USB drive, and photograph the labels.
- Write protection: connect the source through a hardware write blocker so that nothing can be written to it.
- Imaging: in IXImager select the source device and create a full physical image, saved in the ASB container format on a clean, formatted destination drive.
- Verification: IXImager reads back the image and compares it with the source to confirm that every sector was copied.
- Hashing: record the hash (MD5 / SHA-256) of the source and of the image.
- Documentation: note the date and time, examiner, tool and version, hash values, any read errors and the storage location of the image.

Q4. Using hash values to show integrity

A hash value is a fixed-length code calculated from the data. If even one bit changes, the hash becomes completely different. It therefore works as a digital fingerprint.

- Before acquisition: calculate and record the hash of the original drive.
- After acquisition: calculate the hash of the image. If it matches the original, the image is an exact copy.
- Before analysis and before court: calculate the hash of the image again. A match proves it has not been altered since acquisition.
- If the hash differs, the evidence is considered changed or corrupt and the copy must be checked or remade.

Q5. Role of the ASB container

The ASB container (used with IXImager and ILookIX) stores the forensic image in a single protected file together with information about it. It keeps the evidence organized in one place, helps ILookIX load it easily, supports integrity checks, and can keep the image details and examiner notes with the data.

Information that should accompany the evidence: case number and evidence ID; device make, model and serial number; examiner name; date, time and place of acquisition; tool and version; hash values; the write-blocker used; and the chain-of-custody record.

Q6. Data carving and when it is needed

Data carving is the recovery of files from raw data without using the file system. The tool searches for known file signatures, a header at the start (for example a JPEG starts with FF D8 FF) and sometimes a footer at the end, and copies the data in between as a file.

Carving is needed when:

- the file system or file table is damaged or missing,
- the drive has been formatted or re-partitioned,
- files were deleted and their directory entries overwritten, but the data blocks remain,
- the unallocated space must be searched for remains of deleted files.

Limitation: carved files usually lose their name, folder and time stamps, and fragmented files may be incomplete or damaged.

Q7. Date and time problems in digital evidence

- Wrong clocks: the computer's clock may be set incorrectly or changed by the user, so recorded times may not match real time.
- Time zones and daylight saving: the same moment can look different depending on local time or UTC.
- File system differences: NTFS stores times in UTC, while FAT stores local time, so comparing them needs care.
- File operations: copying or moving a file may change its created or modified time. Opening a file may update its accessed time.

How to document: at seizure, compare the system/BIOS clock with a trusted real time and note the difference; record the time zone settings; convert every time stamp to one standard (preferably UTC) in the report; and clearly state any uncertainty, such as "time may be ± 3 minutes".

Q8. Chain-of-custody table (laptop)

Date / Time	Item	Released by	Received by	Purpose	Location / Remarks
16-Oct, 10:00	Laptop (Serial ___)	Seizing officer	Evidence custodian	Seizure and sealing	Office; sealed bag no. 101, photos taken
16-Oct, 12:30	Laptop	Custodian	Courier / examiner	Transport to lab	Seal intact
16-Oct, 14:00	Laptop	Examiner	Evidence locker	Storage	Lab locker, sealed
17-Oct, 09:00	Laptop	Locker	Examiner	Imaging	Write blocker used; hash recorded
17-Oct, 13:00	Laptop	Examiner	Evidence locker	Return after imaging	Original sealed again
18-Oct, 10:00	Forensic image copy	Examiner	Analyst	Examination	Working copy only; hash verified

ASSIGNMENT 2(4)
Recovery of Deleted Evidence from Volume Shadow Copies

Scenario selected: An employee "Meena" is suspected of emailing the confidential file "Salary_Report.xlsx" outside the company on 12 October 2026 and then deleting the file and her browser history. A forensic image of her Windows 10 system is available and contains Volume Shadow Copies.

Q1. Selecting and categorizing digital evidence before detailed analysis

A disk holds a very large number of files, so the examiner must first decide what matters. The steps are:

1. Define the objective: what has to be proven (for example, that the file was copied and deleted by Meena).
2. Limit the data to the relevant users, folders and time period (for example, 1–15 October).
3. Remove irrelevant data: known operating-system and application files are filtered using hash sets such as NSRL.
4. Group the remaining data by type: documents, emails, browser data, system logs, deleted files and so on.
5. Rank by importance and decide the order of analysis.

Q2. Classification scheme

Category	Examples	Use in the case
User-created files	Salary_Report.xlsx, other Office documents, emails	Show the content and the user's work
System / application files	Registry, event logs, Prefetch, browser history database, Jump Lists	Show actions such as program use, logins and web visits
Potentially suspicious files	Unknown .exe, file-wiping or cleaner tools, encrypted archives	Need closer examination, may show an attempt to hide traces
Deleted / recovered artifacts	Recycle Bin items, carved files, files in shadow copies	Show that the user tried to remove evidence
Supporting / contextual evidence	User account list, login times, installed software, USB history	Give background and help prove who and when

Q3. How Event Analysis helps reconstruct activities

Event Analysis gathers time-stamped events from many artifacts (file creation, program execution, logins, USB use, web visits) and displays them together in time order. This lets the examiner see what happened first and what followed. It helps to:

- reconstruct the exact order of actions, such as open file → copy → delete → clear history,
- spot unusual gaps or cleared logs,
- match user activity with the logon session,
- find clusters of activity around a specific time.

Q4. Cloud Analysis and Lead Analysis

Cloud Analysis: examines the traces of cloud services on the computer or phone, such as OneDrive, Google Drive, Dropbox, webmail and sync clients. Questions it helps to answer: Was Salary_Report.xlsx uploaded to a cloud account? Which account was used? When did the upload happen? Which cloud services were installed and used?

Lead Analysis: searches the data for items that may lead to new evidence, such as email addresses, phone numbers, names, IP addresses, URLs and usernames. Questions it helps to answer: Who did Meena contact? Which other accounts or devices are linked? Which new sources should now be examined or requested?

Q5. Volume Shadow Copy analysis

Volume Shadow Copy (VSC) is a Windows feature that creates snapshots of a drive at certain times (for example before an update or in a restore point). Each snapshot keeps the earlier state of files and system data. Even if Meena deleted the file later, an older shadow copy may still hold it.

Investigators use VSC to: recover deleted files, compare an old version of a file with the current one to see what was changed, see old registry and log data that was cleared later, and prove that a file existed at a given date. Comparing several shadow copies also shows how the system changed over time.

Recovery workflow:

1. Verify the hash of the forensic image.
2. List the available shadow copies and note their creation dates.
3. Mount each shadow copy in read-only mode.
4. Compare it with the current volume to find files that are deleted or changed.
5. Export the recovered files and record their hash, path and shadow-copy date.
6. Validate them and add the results to the timeline.

Q6. Hypothetical timeline

Date / Time	Event	Evidence
10-Oct, 09:00	Salary_Report.xlsx created in Meena's Documents folder	$MFT creation time
10-Oct, 17:00	Windows creates a shadow copy (the file is present)	VSC creation time
11-Oct, 11:30	File modified (figures changed)	Modified time; Office recent files
12-Oct, 10:15	File emailed / copied to a USB drive	Email record / USB log
12-Oct, 10:40	File deleted to Recycle Bin and emptied; browser history cleared	Recycle Bin / browser database
12-Oct, 11:00	File no longer found in the live volume	File system
15-Oct (analysis)	File recovered from the 10-Oct shadow copy	VSC analysis

Q7. Validating the recovered artifact before including it in the report

- Verify the image hash and the hash of the recovered file, and record both.
- Open the file and confirm that it is complete and readable.
- Check its metadata: path, owner and times. Confirm they fit the timeline.
- Confirm that it appears in the shadow copy and not in the live volume.
- Cross-check with other artifacts such as Recycle Bin entries, Jump Lists, email logs or USB records.
- Repeat the recovery using a second tool or a second examiner.
- Document the steps, tool versions and results so the work can be repeated.

Q8. Possible sources of error and alternative explanations

- Clock and time zone: wrong system clock or time-zone differences can give wrong times.
- Shadow copy time: the time of a shadow copy is when the snapshot was made, not when the user acted.
- Other causes of deletion: Windows cleanup, an antivirus, a different user or an automatic synchronization could have deleted the file.
- Timestamp changes: updates, backups, copying or antivirus scans can alter file times.
- Partial recovery: parts of a deleted file may be overwritten, so the recovered data may be incomplete or damaged.
- Missing shadow copies: copies can be deleted or may not exist for the relevant period, so absence of a file does not prove it never existed.

ASSIGNMENT 2(5)
Multi-Source Investigation of a Suspected Cyber Incident

Scenario selected: A student account "stu123" was used to log in to the college online exam portal and change marks on 14 October 2026 at about 21:00. Evidence is available (with proper authorization) from a lab Windows PC, its browser, email, messaging records, a mobile phone and cloud/Internet logs.

Q1. Potential evidence from Windows and other operating systems

- Windows: registry (user activity, USB devices, installed programs), event logs (logon and logoff), user profiles, Prefetch (program execution), recent files, Jump Lists, Recycle Bin and the page file.
- Linux: authentication logs, shell history, system logs, cron jobs and user home directories.
- macOS: unified logs, user preferences, Spotlight and Time Machine records.
- Memory and network data: running processes, active connections and cached passwords, if the system was running.

Q2. What browser artifacts reveal, and why they need careful interpretation

Browsers keep history, cache, cookies, download lists, saved passwords, form data, search terms and bookmarks. These show which websites were visited and when, what was downloaded, which accounts were logged in, and what the user searched for.

They must be interpreted carefully because:

- a visit in the history does not prove who was sitting at the keyboard,
- private/incognito mode leaves little data, and users can clear history,
- pages can load automatically (pop-ups, redirects, embedded content) and appear as visits,
- time stamps depend on the computer's clock and time zone,
- a shared computer can have several browser profiles.

Q3. Email and messaging evidence and its correlation

Types of evidence:

- Email: headers (sender, receiver, route, IP address, time), body text, attachments, sent and deleted items, and mailbox files such as PST/OST.
- Messaging apps: chat text, contacts, shared files and images, call logs, group membership and time stamps.

Correlation: compare the time of an email or chat with browser visits, file creation and login times; match the IP address in the email header with portal or server logs; match the hash of an attachment with a file found on the computer; and link phone numbers or email addresses to the accounts used in the incident.

Q4. Forensic considerations for mobile phone examination

- Isolate the phone from the network (airplane mode or a Faraday bag) to stop remote wipe or new messages.
- Document the phone's state, make, model, IMEI and photograph it. Keep it charged.
- Use legal authority and, where needed, the PIN or unlock method.
- Choose the acquisition type (logical, file-system or physical) that the device and tool support, and record hash values.
- Examine call logs, SMS, app data, photos, location data and browser history.

Challenges: screen locks and full-disk encryption, many device models and operating-system versions, rapidly changing apps, auto-deleting messages, cloud sync that changes data, and data that is hard to read.

Q5. Internet and cloud-based evidence sources

- Server and web logs of the exam portal: IP addresses, login times, actions performed.
- ISP and IP address records that show who held an IP address at a given time (needs a legal request).
- Cloud storage and accounts: Google Drive, OneDrive, iCloud files, login history, device lists and backups.
- Email and social-media providers: account details, logs and messages (through legal process).
- College firewall, proxy and Wi-Fi logs that show the device, MAC address and websites visited.

Q6. Cross-source correlation table

Source	Artifact	Time	What it shows
Computer	Windows event log: stu123 logon on lab PC	20:55	The account was used on the lab computer
Browser	Visit to the exam portal admin page	21:00	The portal was accessed from that PC
Email	Password-reset email received by stu123	20:50	The password was changed shortly before the login
Mobile	Chat message: “done, check marks”	21:15	Possible discussion of the act
Cloud / server	Portal log: marks changed from IP 10.1.5.22	21:02	IP matches the lab PC
Firewall	Lab PC connection to the portal at 21:00	21:00	Confirms the network connection

Q7. Major challenges when evidence is spread across devices and cloud

1. Large volume of data: many devices and services create a huge amount of data to collect and review.
2. Jurisdiction and law: cloud data may be stored in other countries and needs legal requests that take time.
3. Limited access: the cloud provider controls the data, so a full forensic image of cloud data is often not possible.
4. Different formats and tools: each device and app stores data differently and needs different tools.
5. Encryption and anti-forensics: encrypted devices, secure-delete tools and auto-delete apps hide evidence.
6. Time synchronization: devices use different clocks and time zones, which makes the timeline difficult.
7. Chain of custody and integrity: it is harder to prove that online data was not changed.
8. Privacy: personal data of other people is mixed in with the evidence and must be protected.

Q8. Final investigation conclusion

Type	Findings
Direct evidence	Portal log shows marks changed using stu123 from the lab PC’s IP at 21:02. Windows event log shows stu123 logon on the PC at 20:55.
Corroborating evidence	Browser history shows the admin page visit at 21:00; password-reset email at 20:50; chat message at 21:15; firewall log confirms the connection.
Unresolved questions	Who was physically using the PC? Was the password shared or stolen? Does the phone belong to the same person? Is CCTV or lab-entry data available? Are all cloud records obtained?

Thank you